

# Literature Status Note for arXiv:1601.07993

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**Status.** `literature_already_answered`. This note records a separate later paper's exact answer; it is not a new proof from this run.

## Source Question

The source is K. R. Davidson, A. Dor-On, O. M. Shalit, and B. Solel, *Dilations, inclusions of matrix convex sets, and completely positive maps*, arXiv:1601.07993 [1].

For a convex body  $K \subseteq \mathbb{R}^d$ , let

$$\theta(K) = \inf\{C > 0 : \mathcal{W}^{\max}(K) \subseteq C\mathcal{W}^{\min}(K)\}.$$

On page 35, Remark 7.21, the source says that it is not known whether the factor  $d$  can be improved for a particular  $d$ -dimensional convex set and asks, in particular, whether  $d$  is optimal for  $K = [-1, 1]^d$ .

## Supporting Answer

The supporting paper is B. Passer, O. M. Shalit, and B. Solel, *Minimal and maximal matrix convex sets*, arXiv:1706.05654 [2]. It cites the source as reference [6] and explicitly identifies the formerly open cube problem on page 2.

On page 29 it proves

$$\mathcal{W}^{\max}([-1, 1]^d) \subseteq \sqrt{d}\mathcal{W}^{\min}([-1, 1]^d)$$

and states that the factor  $\sqrt{d}$  is optimal. Hence

$$\boxed{\theta([-1, 1]^d) = \sqrt{d}.}$$

Theorem 6.9 on page 32 gives the broader sharp formula

$$\theta(B_{p,d}) = d^{1-|1/p-1/2|}, \quad 1 \leq p \leq \infty,$$

whose  $p = \infty$  case is exactly the cube.

Thus the source's factor  $d$  is not optimal for the cube when  $d > 1$ ; the exact constant is  $\sqrt{d}$ . Equivalently, every  $d$ -tuple of self-adjoint contractions has a commuting self-adjoint dilation with coordinate norms at most  $\sqrt{d}$ , and the uniform factor cannot be decreased.

## Verification

The source question was checked in the local source PDF at page 35, Remark 7.21. The supporting identification was checked in the local answer PDF at pages 2, 29, and 32. The answer paper explicitly cites the source and describes the cube constant as previously unknown, so this belongs in the direct later-literature bucket rather than an agent-derived result bucket.

## References

- [1] K. R. Davidson, A. Dor-On, O. M. Shalit, and B. Solel, *Dilations, inclusions of matrix convex sets, and completely positive maps*, arXiv:1601.07993.
- [2] B. Passer, O. M. Shalit, and B. Solel, *Minimal and maximal matrix convex sets*, arXiv:1706.05654.